Activity 14 — ShopSmart: Predicting Customer Churn

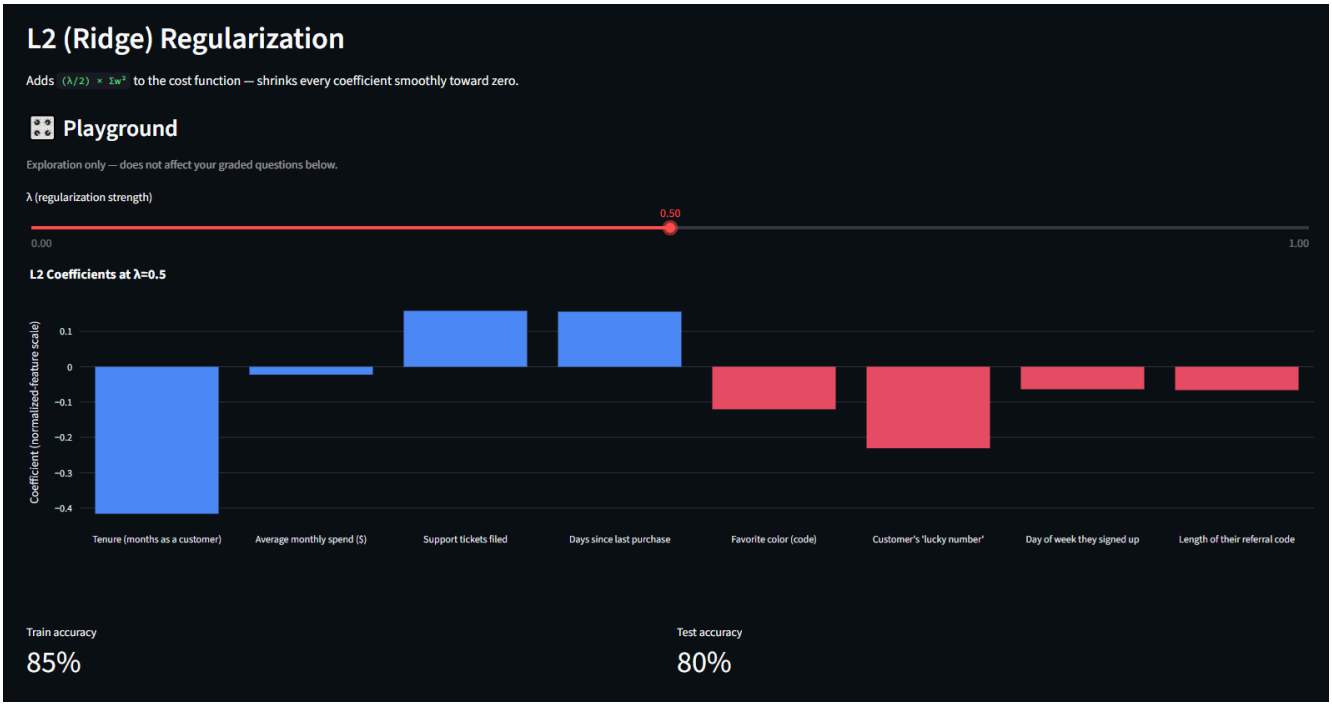
Labeled screenshots · Session 24 · Diego Kamil Olea Ruiz · Artificial Intelligence

A — The Overfitting Problem: plain model

Supports questions 1 and 7. Without regularization the model hits 100% train accuracy but only 77% on test. Train loss is 0.022 and test loss is 2.887, so it memorized the training customers instead of learning the real pattern.

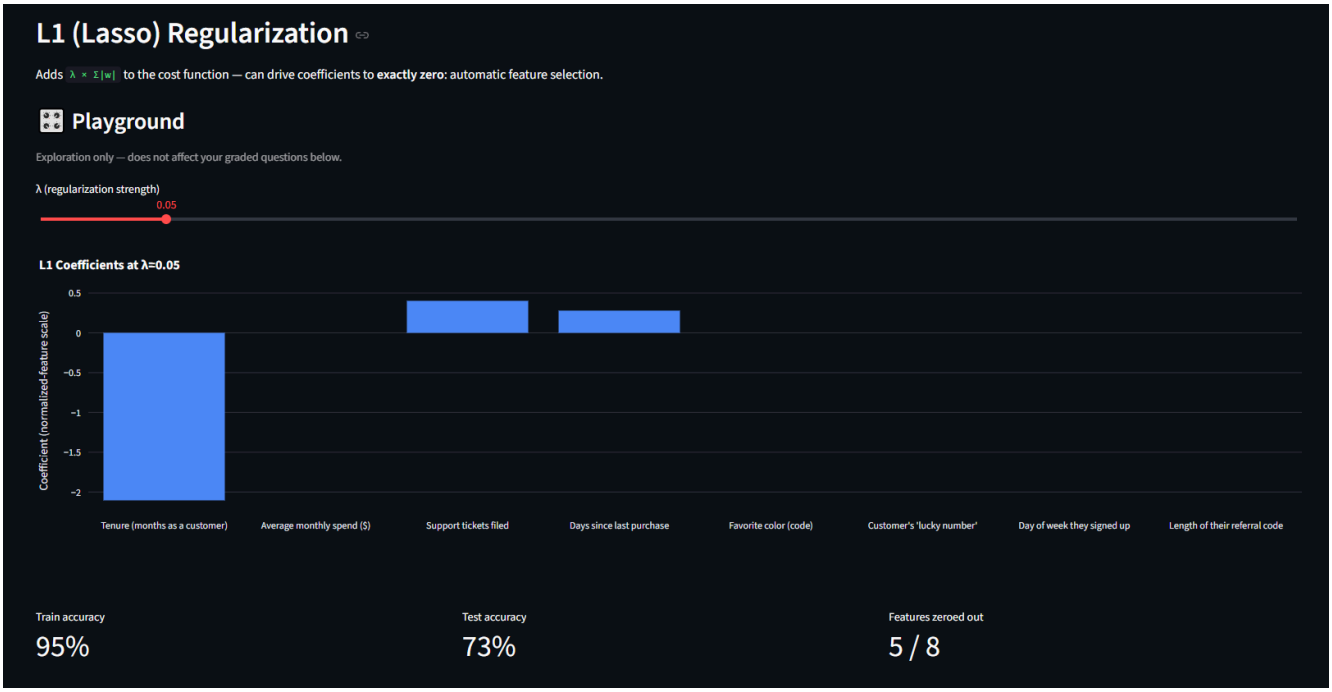
B — L2 playground at a large λ

Supports question 6. At λ = 0.5 all 8 bars shrank a lot but none of them reached zero. Train accuracy went down to 85% and test stayed at 80%.



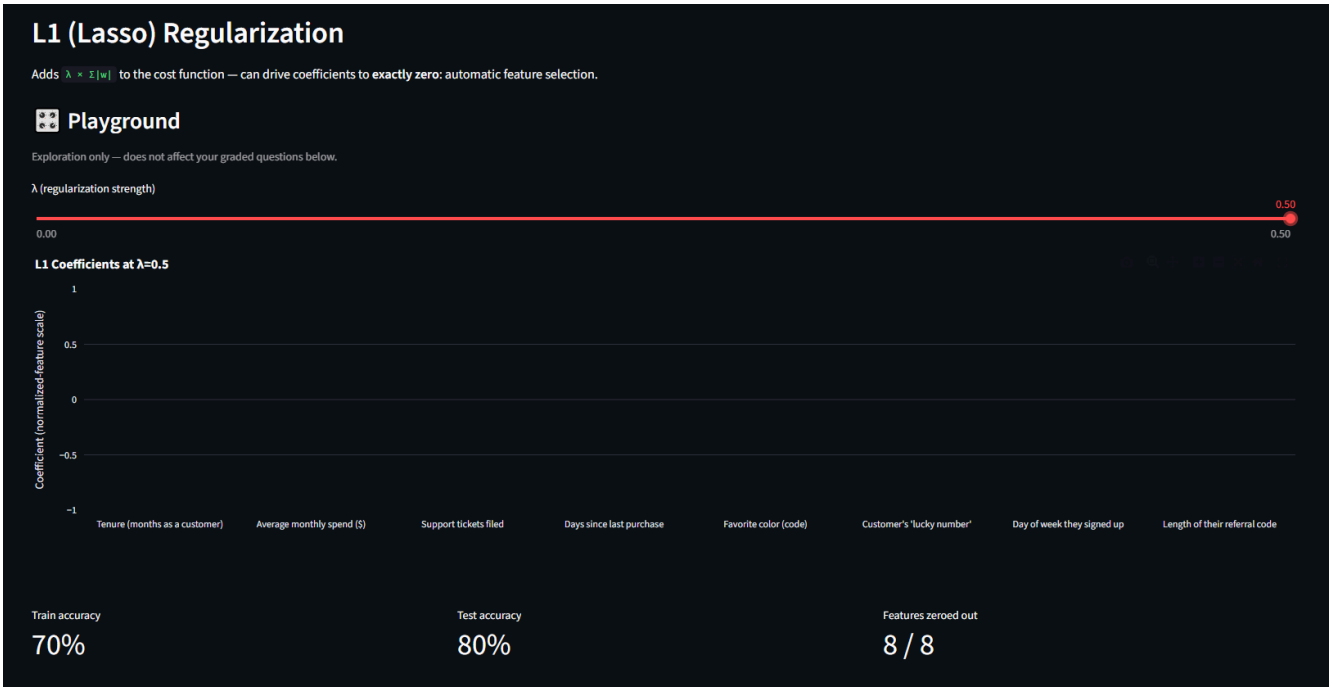
C — L1 playground with features zeroed out

Supports question 6. At λ = 0.05 L1 already zeroed out 5 of 8 features, all 4 noise ones plus average monthly spend. Only tenure, support tickets and days since last purchase are left.



D — L1 playground at λ = 0.5

Supports question 6. Pushing L1 to λ = 0.5 zeroes out all 8 features, so the model only predicts from the bias. That is where L1 goes too far.



E — Canonical L2 Model (λ = 0.05)

Supports questions 2 and 3. The graded L2 model gets 95% train and 83% test accuracy. It is 5 points lower on train but 7 points higher on test than the plain model, so it does better on new customers.

F — Canonical L1 Model (λ = 0.05)

Supports questions 4 and 5. The graded L1 model gets 95% train and 73% test accuracy, with 5 of 8 features zeroed out: average monthly spend, favorite color, lucky number, signup weekday and referral code length.